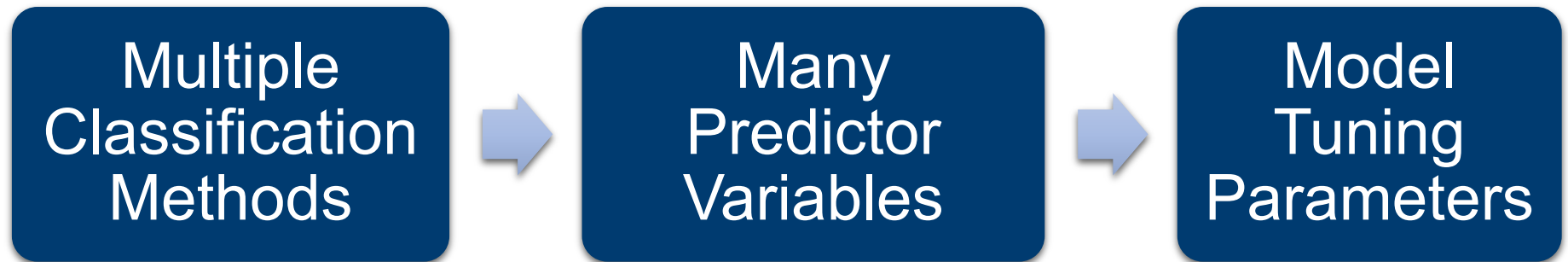


Data Mining for Business Analytics

MSBA 511

Evaluating Classifiers

Why Evaluate?



Need to **evaluate** performance to help choose the optimal method and parameters

Overfitting Revisited

We learned last week that Decision Trees are very susceptible to overfitting by building very complex trees with 100% purity in each leaf.

This problem is not unique to Decision Trees as all classification models can suffer from overfitting.

Some causes:

- Too many features (Curse of dimensionality)
- Excessive model complexity
- Data leakage

Consequence:

- Deployed model will not generalize well with new data.

How to Address this Problem?

Solution: Partition the data



The diagram consists of two large, right-pointing chevrons. The left chevron is teal and contains the text 'Training Data'. The right chevron is light blue and contains the text 'Validation Data'. Below each chevron is a descriptive sentence. The chevrons are positioned side-by-side, suggesting a flow or comparison between the two data partitions.

**Training
Data**

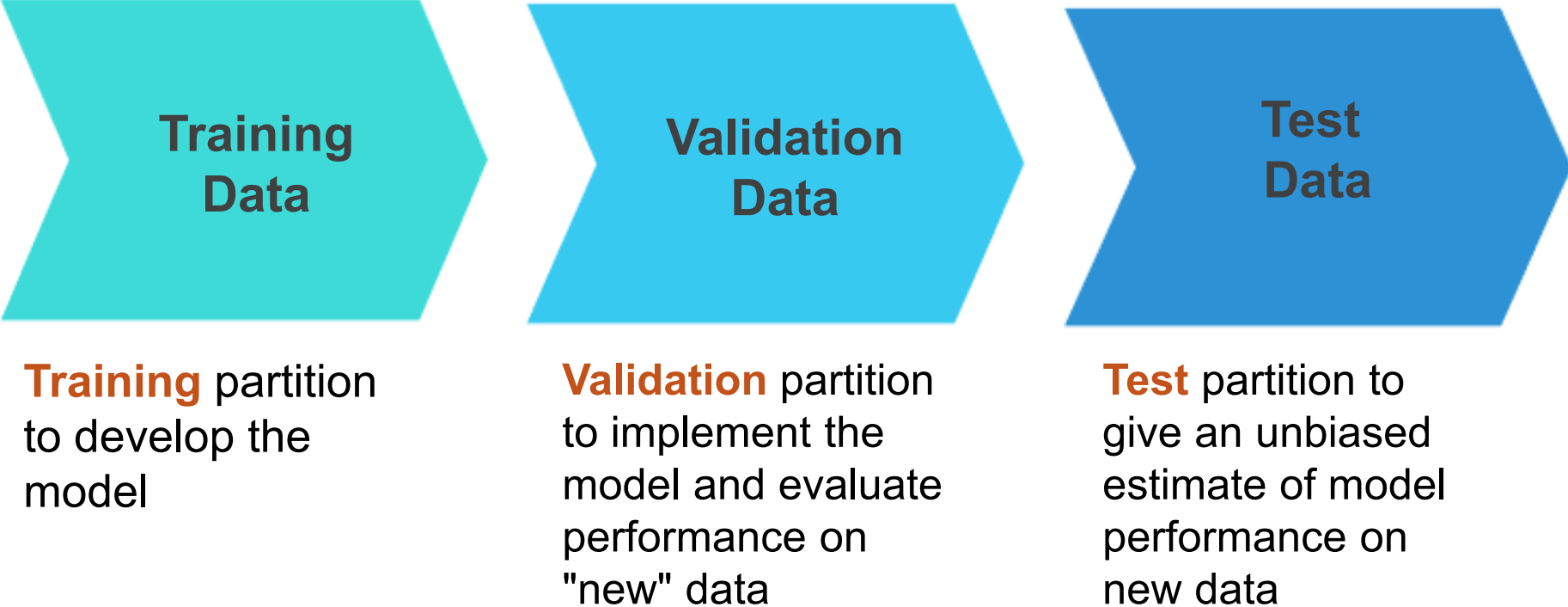
Training partition to develop the model

**Validation
Data**

Validation partition to implement the model and evaluate performance on "new" data

How to Address this Problem?

Any concerns with this approach?



**Training
Data**

Training partition
to develop the
model

**Validation
Data**

Validation partition
to implement the
model and evaluate
performance on
"new" data

**Test
Data**

Test partition to
give an unbiased
estimate of model
performance on
new data



Assessing multiple models and/or tuning models using the validation data to make parameter choices can lead to overfitting the validation data.

"Test" Partition Terminology

On the prior slide and in the textbook, the **"test"** partition is a third partition to provide an unbiased assessment of model performance.

More generally in data science, and in Python syntax, **"test"** refers to the partition set aside from the training partition.

Cross Validation

So, can we still have a validation partition?

Yes, Many model tuning techniques use repeated partitioning of the training data known as cross-validation ("cv") to identify optimal parameters and to compare different models.

k-fold cross-validation

- For each fold, set aside $\frac{1}{k}$ of data as validation
- Use remainder of data for training
- The validation folds are non-overlapping

Accuracy Measure

$$\text{Accuracy Rate} = \frac{\text{Number of Correct Classifications}}{\text{Number of Records in Dataset}}$$

Income (K)	Credit Score	Loan Amount (K)	Age	Default (Actual)	Default (Predicted)
50	700	10	35	0	0 ✓
30	600	15	28	1	1 ✓
80	750	20	45	0	0 ✓
25	580	12	24	1	0 ✗
60	720	18	40	0	0 ✓

**In this loan default example,
Accuracy is 4 out of 5 or 80%**

Is Good Accuracy Always Good?

Example: typical response rate for a mailing campaign is 1%

- Two models
 - Model A accurately classifies 90% of instances in the test set.
 - Model B accurately classifies 99% of instances in the test set.

Which model would you pick?

- Model B predicts that no customer will respond so has a classification accuracy of 99%.
 - Misclassifies all respondents - **Very accurate, but useless.**

Accuracy alone is not a sufficient metric for imbalanced classification problems.

Naïve Classifier Benchmark

On the prior slide, Model B could be used as **Naïve classifier** benchmark for accuracy by simply classifying all records as belonging to the most prevalent class. Involves making predictions using a straightforward, rule-based approach without leveraging complex patterns in the data.

Sets a ***baseline accuracy*** that any model should perform significantly better to be considered useful.



The primary exception is when the goal is to predict high-value but rare outcomes (fraud detection, loan default, disease diagnosis) where accuracy lower than a benchmark could still be useful.

Confusion Matrix

- A **confusion matrix** records the source of error:
 - **False positives** and **False negatives**

$$Accuracy = \frac{TP + TN}{P + N}$$

Continuing with the mailing campaign and suppose **1,000** mailers are sent out. **What is accuracy?**

Actual class

Predicted class		
	Positive	Negative
Positive (P)	True positive (TP)	False negative (FN)
Negative (N)	False positive (FP)	True negative (TN)

Actual class

Predicted class		
	Respond	Do not Respond
Respond	0	10
Do not Respond	0	990



In the textbook as well as sklearn metrics, the positive class is shown in the second column and bottom row of the confusion matrix.

Cutoff Value for Classification

Most algorithms classify via a 2-step process:

For each record:

1. Compute **probability of belonging to positive class**
2. Compare to cutoff value, and classify accordingly

Default cutoff value is 0.50

If ≥ 0.50 , classify as “1”

If < 0.50 , classify as “0”

- Can use different cutoff values
- Typically, error rate is lowest for cutoff = 0.50

Cutoff Table Example

Actual	Predicted Probability	Actual	Predicted Probability
owner	0.9959	owner	<u>0.5055</u>
owner	0.9875	nonowner	0.4713
owner	0.9844	nonowner	0.3371
owner	0.9804	owner	0.2179
owner	0.9481	nonowner	0.1992
owner	0.8892	nonowner	0.1494
owner	0.8476	nonowner	0.0479
nonowner	<u>0.7628</u>	nonowner	<u>0.0383</u>
owner	0.7069	nonowner	0.0248
owner	0.6807	nonowner	0.0218
owner	0.6563	nonowner	0.0161
nonowner	0.6224	nonowner	0.0031

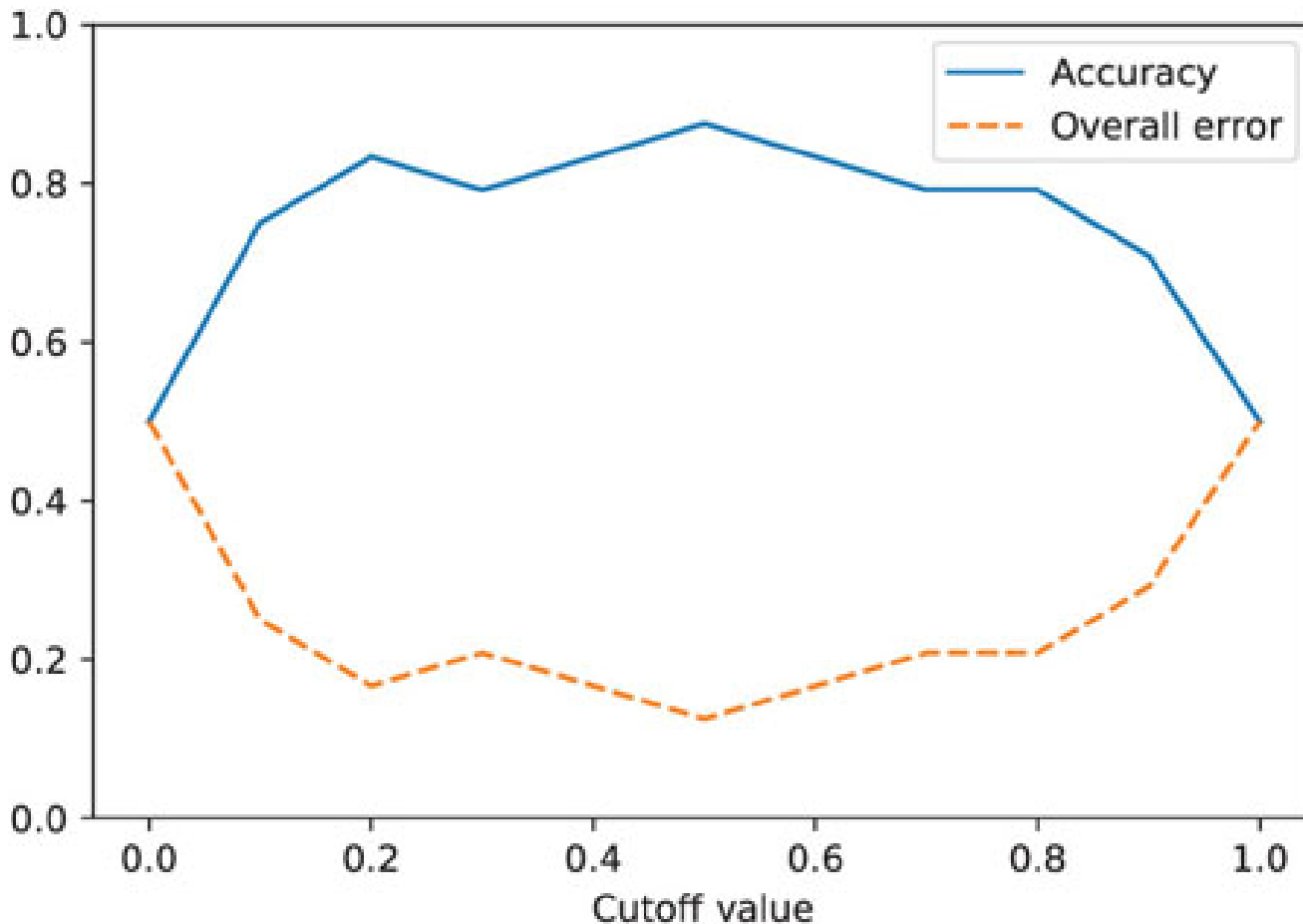
Standard cutoff of 0.5 results in 0.875 accuracy

Higher cutoff of 0.75 reduces accuracy to .75

Lower cutoff of 0.25 reduces accuracy to .792

Visualizing Cutoff Value Tradeoff

Adjusting the **cutoff value** in a classification model impacts the tradeoff between accuracy and misclassification error.



Precision

$$Precision = \frac{TP}{TP + FP}$$

Also referred to as
Positive Predictive
Value (PPV)

**Actual
class**

Predicted class

	Positive	Negative
Positive (P)	True positive (TP)	False negative (FN)
Negative (N)	False positive (FP)	True negative (TN)

False Discovery
Rate (FDR) is
1 - PPV

Example: In spam detection, how many flagged emails are spam?

False Omission Rate Rate (**FOR**)

$$FOR = \frac{FN}{TN + FN}$$

		Predicted class	
		Positive	Negative
Actual class	Positive (P)	True positive (TP)	False negative (FN)
	Negative (N)	False positive (FP)	True negative (TN)

Example: With job applications, how many rejected candidates were qualified?

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

Also referred to as True Positive Rate (TPR), Sensitivity or "Hit Rate"

Type II Error or False Negative Rate (FNR) is $1 - \text{TPR}$ or "Miss Rate"

		Predicted class	
		Positive	Negative
Actual class	Positive (P)	True positive (TP)	False negative (FN)
	Negative (N)	False positive (FP)	True negative (TN)

Example: In medical diagnostics, how many cancer cases are detected?

False Positive Rate (**FPR**)

$$FPR = \frac{FP}{(FP + TN)}$$

This is Type I Error
or "False Alarm"

**Actual
class**

Predicted class

	Positive	Negative
Positive (P)	True positive (TP)	False negative (FN)
Negative (N)	False positive (FP)	True negative (TN)

TNR or
Specificity is
 $1 - FPR$

Example: In security screening, how many non-threats were incorrectly flagged?

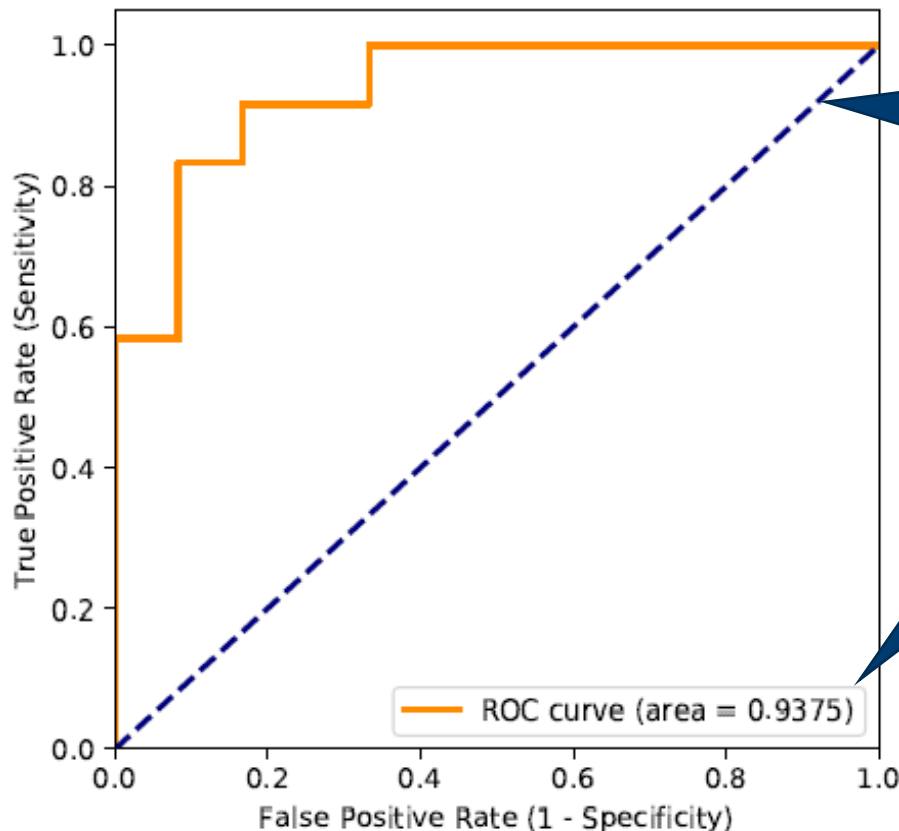
Receiver Operating Characteristic (ROC)

The term **Receiver Operating Characteristic (ROC)** originates from World War II radar signal detection. Military engineers and operators used it to evaluate the performance of radar systems in distinguishing enemy aircraft (**signal**) from background noise (**non-signal**).

This concept is now a very popular method to assess the performance of binary classification models by measuring their ability to differentiate between positive and negative classes.

ROC Curve

Visualization that shows the trade-off between the TPR (Recall/Sensitivity) and FPR (1 – TNR/Specificity) at various classification thresholds.



Diagonal line represents a random classifier, and predictions are no better than 50-50 chance (AUC = 0.5)

AUC (area under the curve) is a common metric to summarize the ROC curve and ranges from 1 to 0.5

Why All of These Metrics?

Choosing the right evaluation metric is crucial because different metrics reflect different perspectives on model performance.

- Metrics such as recall (sensitivity) and TNR (specificity) are from the business perspective (e.g., company classifies customers, flags potential fraudulent transactions, hospital classifies patients).
- Metrics such as FDR and FOR are from the perspective of the entity being classified and how likely to belong in the predicted class.

The choice of metric should align with the **business objective** and the cost (opportunity) of errors.

Confusion Matrix Evaluation Example

Below shows the performance of two classifiers.

Which one is better based on accuracy, precision, and recall?

	Respond	Do not respond
Respond	5	5
Do not Respond	40	950

$$\text{Accuracy} = \frac{(5 + 950)}{1000} = \mathbf{95.5\%}$$

$$\text{Precision} = \frac{5}{(5 + 40)} = \mathbf{11.1\%}$$

$$\text{Recall} = \frac{5}{(5 + 5)} = \mathbf{50.0\%}$$

	Respond	Do not respond
Respond	10	0
Do not Respond	90	900

$$\text{Accuracy} = \frac{(10 + 900)}{1000} = \mathbf{91.0\%}$$

$$\text{Precision} = \frac{10}{(10 + 90)} = \mathbf{10.0\%}$$

$$\text{Recall} = \frac{10}{(10 + 0)} = \mathbf{100.0\%}$$

Cost Based Evaluation

The business objective influences your choice. Suppose the cost of mailing to a non-responder is **\$1**, and (net) lost revenue of not mailing to a responder is **\$20**.

Now from a cost perspective, which one is better?

	Respond	Do not respond
Respond	5	5
Do not Respond	40	950

$$\text{Cost} = (5 * 20) + (40 * 1) = \text{\$140}$$

	Respond	Do not respond
Respond	10	0
Do not Respond	90	900

$$\text{Cost} = (0 * 20) + (90 * 1) = \text{\$90}$$

Profit Based Evaluation

Now suppose on average for each responder, upon receiving a mailer, contributes \$21 of revenue and the mailing cost \$1 per customer.

Now from a profit perspective, which one is better?

	Respond	Do not respond
Respond	5	5
Do not Respond	40	950

$$\text{Profit} = (5 * 21) - (45 * 1) = \$60$$

	Respond	Do not respond
Respond	10	0
Do not Respond	90	900

$$\text{Profit} = (10 * 21) - (100 * 1) = \$110$$

Lift and Gains

Evaluates how well a model identifies the most important class.

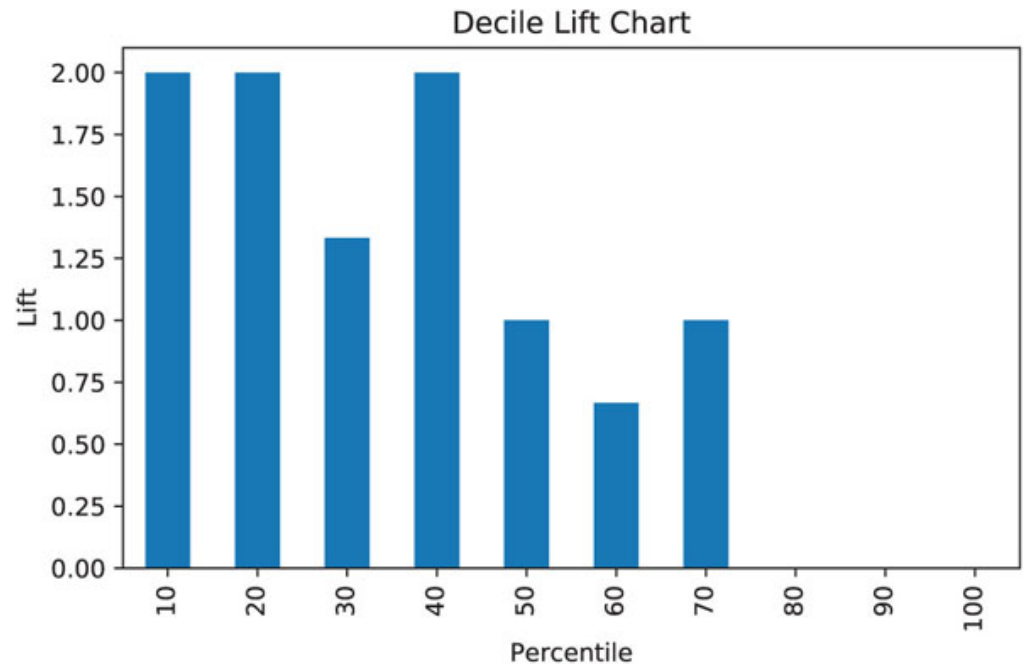
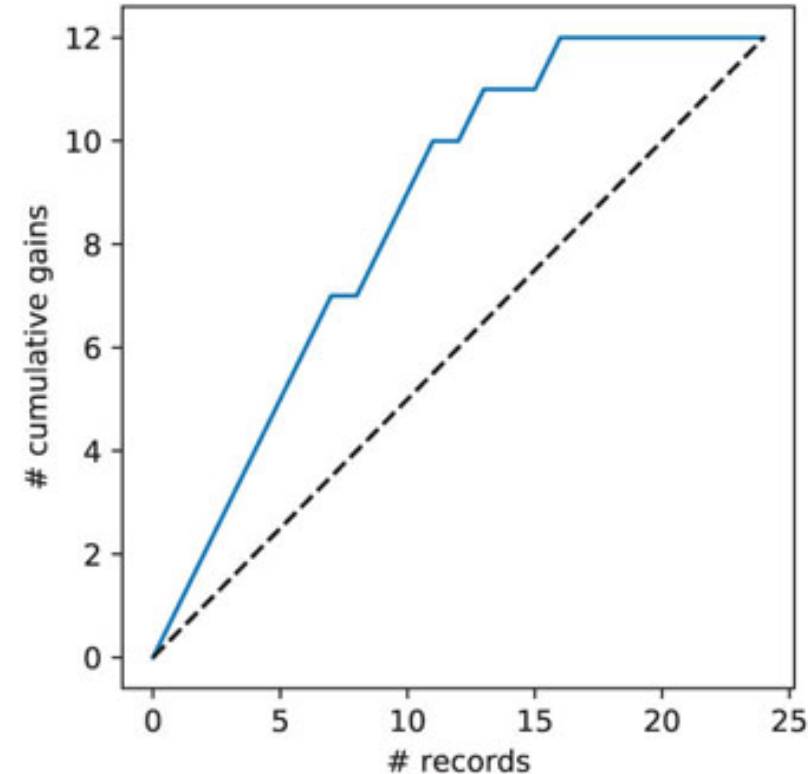
1. Sort records by **predicted probability** of belonging to the important class.
2. Move down the list, noting **actual** class and **cumulative actual class**.

Rank	Predicted Probability	Actual Class	Cumulative Actual Class
1	0.995	1	1
2	0.988	1	2
3	0.984	1	3
4	0.980	1	4
5	0.948	1	5
6	0.889	1	6
7	0.847	1	7
8	0.763	0	7
9	0.707	1	8
10	0.681	1	9
11	0.656	1	10
12	0.622	0	10

Cumulative Gains and Decile Lift Chart

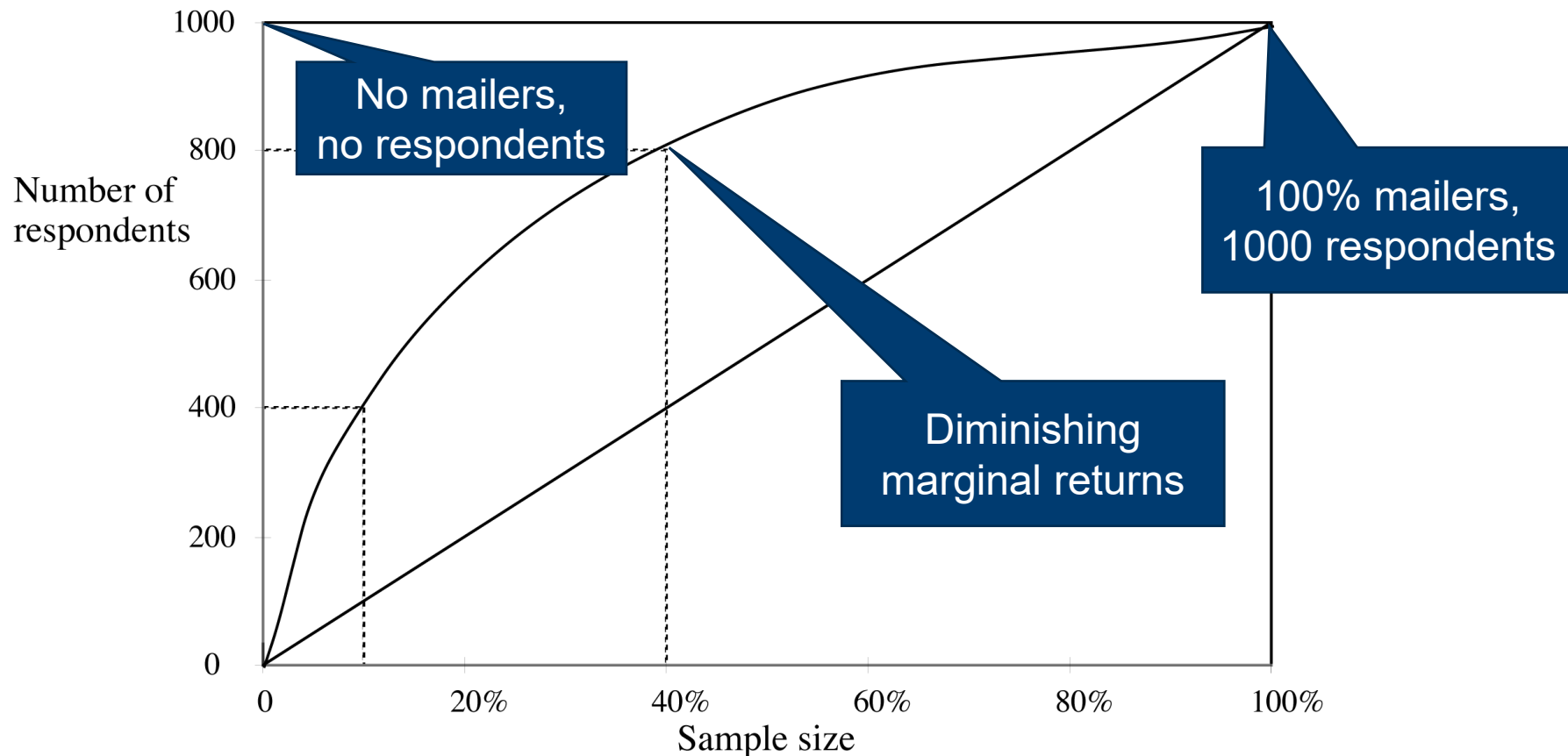
Plot the number of **cumulative gains** as a function of the number of records.

Plot the **lift** over a random 50% chance model by number of record percentiles.



Lift Chart Example

- Plot the number of **cumulative** respondents as a function of the number of mailings.



Summary

Accuracy – Measures overall correctness: Best for balanced datasets, but misleading for imbalanced classes.

Precision – Focuses on positive predictions: Useful when false positives are costly (e.g., fraud detection, spam filtering).

Recall (Sensitivity) – Measures how many actual positives were identified: Critical when missing positives is risky (e.g., medical diagnosis, security alerts).

ROC Curve & AUC – Evaluates model performance at different thresholds.

- **Higher AUC** = Better discrimination between classes.
- Helps find the optimal balance between true positives and false positives.

Lift & Gains Charts – Assess how well the model **ranks and prioritizes** high-value cases.

- **Lift** shows improvement over random selection (e.g., marketing campaigns).
- **Gains** visualize how well the model captures true positives early.

The best evaluation metric depends on the business objective, whether prioritizing overall accuracy, reducing false positives, capturing rare events, or optimizing resource allocation.